Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

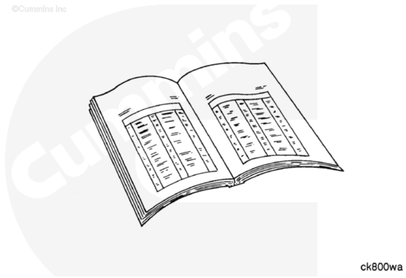
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries or air supply line to the air starter to prevent accidental engine starting. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html).
- Remove the coolant filter. Refer to Procedure 008-006 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-006-tr.html).

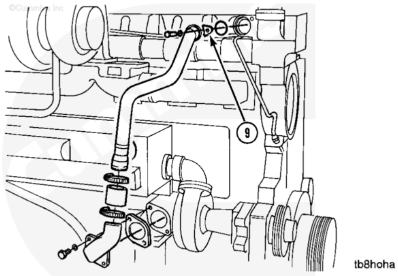


Remove

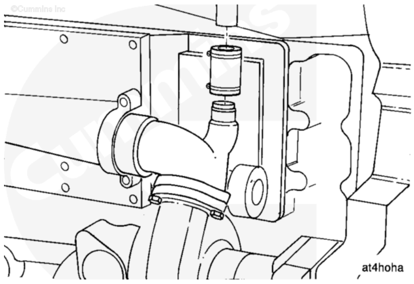
Remove the water bypass tube clamp (9).

Loosen both hose clamps.

Remove the water bypass tube.



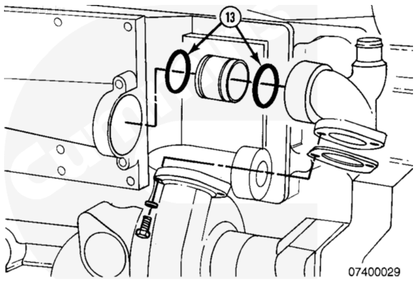
Disconnect the aftercooler supply hose from the water pump outlet connection.



Remove the water pump outlet connection assembly.

Remove the water transfer tube from the water connection.

Remove and discard the two o-rings (13) and the gasket.



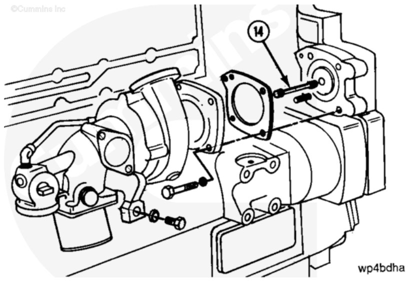
Remove the capscrew from the water pump support bracket.

Remove the three capscrews and the nut from the water pump mounting flange.

Remove the water pump and gasket.

Discard the gasket.

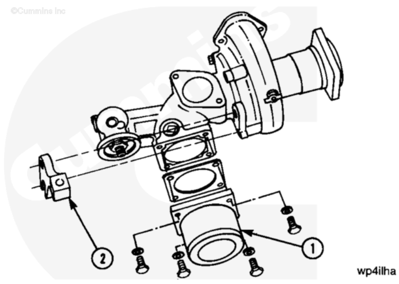
Remove the water pump drive shaft (14).



Disassemble

Remove the water inlet connection (1) and gasket.

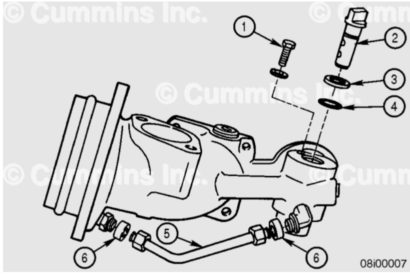
Remove the support bracket (2).



Remove the following:

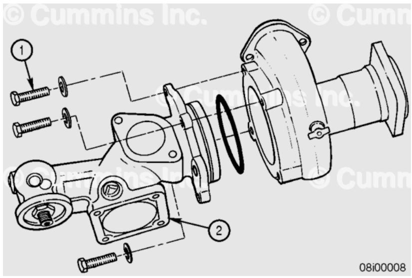
1. Shutoff valve capscrew
2. Water shutoff valve
3. Washer
4. O-ring
5. Water transfer tube (loosen the flare nuts at each end)
6. Grommets.

Discard the o-ring and grommets.



Remove the remaining capscrews (1), inlet housing (2), and o-ring.

Discard the o-ring.



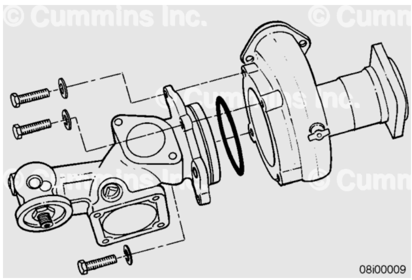
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

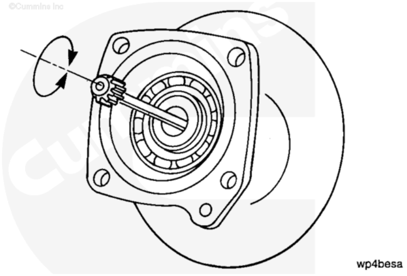


Clean the parts with solvent, Part Number 3824421, or equivalent.

Dry with compressed air.

Rotate the shaft and inspect for rough or damaged bearings.

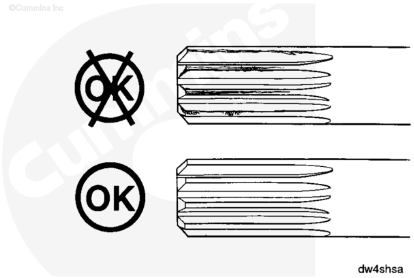
If the bearings are rough or damaged, the water pump **must** be replaced.



Inspect the drive shaft for wear.

If the drive shaft splines are worn, check the female splines in the pump and in the water pump drive.

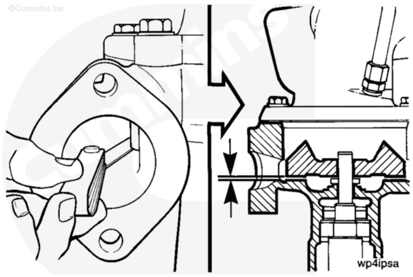
Replace worn parts as necessary.



Use a feeler gauge to measure the impeller to water pump body clearance.

Impeller to Pump Body Clearance		
mm		in
0.84	MIN	0.033
1.02	MAX	0.040

If the pump is **not** within specifications, it **must** be replaced.



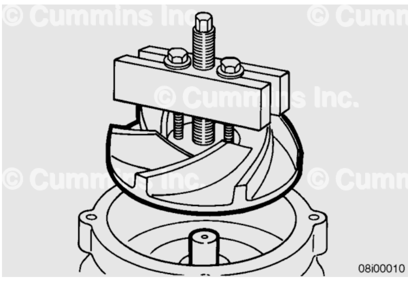
⚠ CAUTION ⚠

The jack screw in the puller must pass easily through the impeller bore to reduce the possibility of damage.

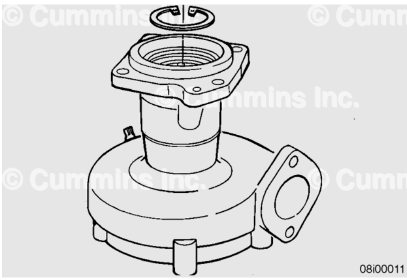
Remove the impeller with standard puller, Part Number ST-647, or equivalent.

Clean the impeller.

Check the impeller for damage.



Remove the large retaining ring.

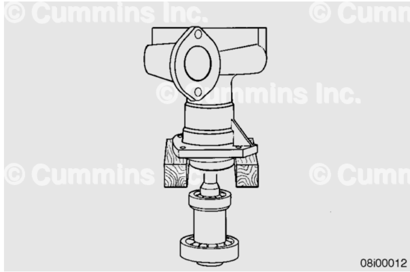


Support the water pump body on an arbor press.

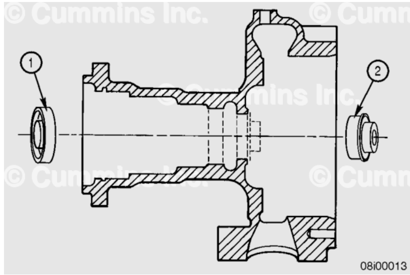
Press the impeller end of the shaft.

Remove the bearing and shaft assembly.

Discard the water pump seal seat.



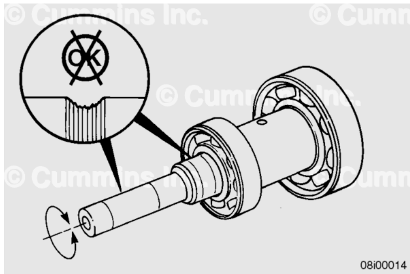
Remove the water seal (1) and oil seal (2).



Check the shaft for wear in the seal areas.

Spin the bearings to check for roughness.

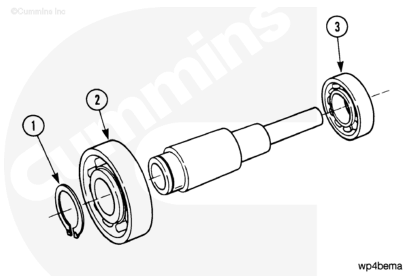
Replace the bearings and/or shaft if damage is found.



Use a water pump bearing separator, Part Number 3375326, or equivalent.

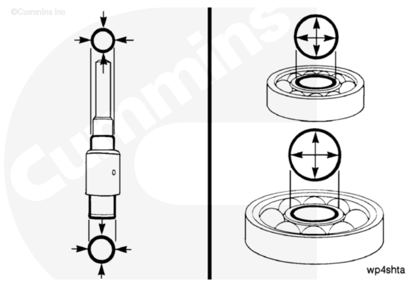
Disassemble the parts.

1. Retaining ring
2. Front bearing
3. Rear bearing.



Measure the shaft outside diameter (O.D.) and the bearing inside diameter (I.D.).

Compare the differences.

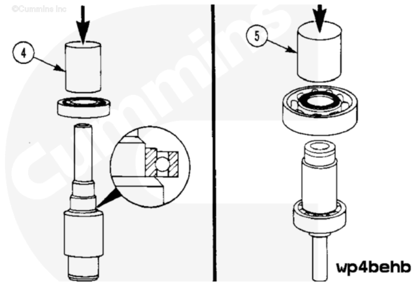


Measurements		
	mm	in
Maximum Clearance	0.003	0.0001

Measurements		
	mm	in
Maximum Interfereance	0.018	0.0007

⚠ CAUTION ⚠

The mandrel must make contact on the inner race of the bearing to reduce the possibility of damage.

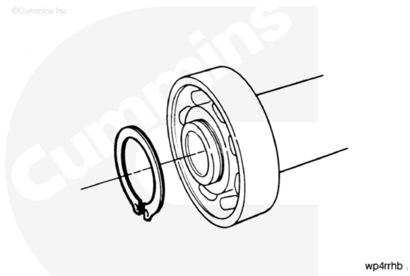


Use a water pump bearing mandrel, Part Number ST-658, or equivalent (4).

Support the shaft on an arbor press.

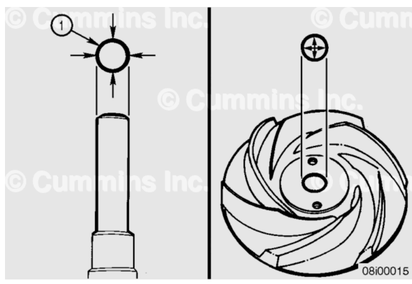
Install the bearings.

Install the retaining ring.



Use the following formula to determine the press fit.

Press Fit: Shaft O.D. minus impeller I.D.

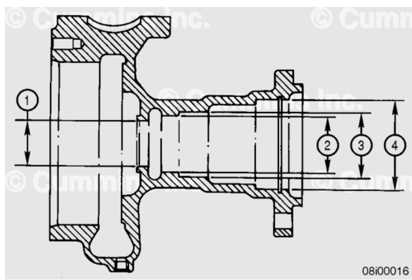


Impeller to Shaft Press Fit		
mm		in
0.03	MIN	0.001
0.07	MAX	0.003

Remove the impeller with standard puller, Part Number ST-647, or equivalent.

Clean the impeller.

Check the impeller for damage.



Housing Bore I.D.			
	mm		in

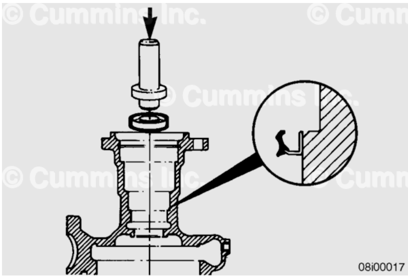
Water Seal (1)	36.45	MIN	1.435
	36.47	MAX	1.436
Oil Seal (2)	44.43	MIN	1.749
	44.48	MAX	1.751
Rear Bearing (3)	51.996	MIN	2.0477
	52.012	MAX	2.0471
Front Bearing (4)	71.996	MIN	2.8345
	72.011	MAX	2.8351

Assemble

The oil seal **must** be installed facing downward.

Install the seal with water pump bearing mandrel, Part Number 3375320, or equivalent.

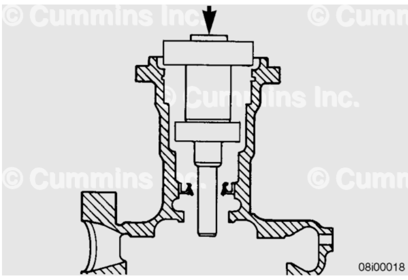
The seal **must** be no more than 0.51 mm [0.020 in] below the top of the step in the body.



Support the water pump body in an arbor press.

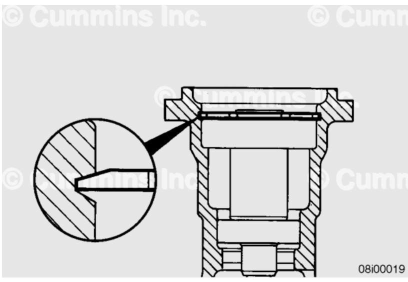
Install the shaft/bearing assembly into the housing.

Press the shaft into the housing with water pump bearing mandrel, Part Number 3375318, or equivalent.



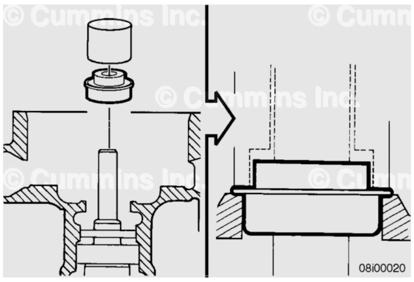
The large retaining ring **must** be installed with the bevel facing away from the bearing.

Install the large retaining ring.



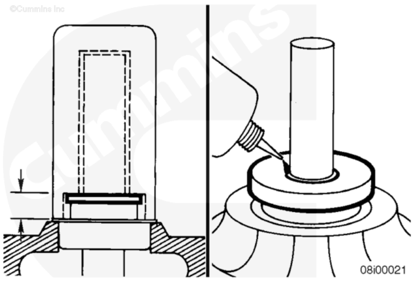
Install the water pump seal with the water pump seal driver, Part Number 3823815, or equivalent.

Be sure the shoulder on the seal touches the housing.



⚠ CAUTION ⚠

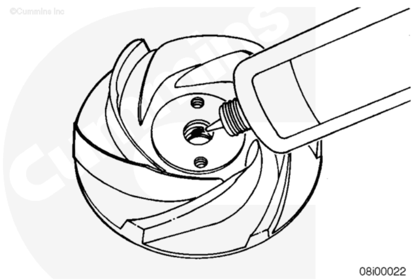
More than one drop Loctite® 290 will glue the faces of the seal and seat together. It can result in damage to the seal.



Seat Dimension		
mm		in
10.92	MIN	0.430
11.68	MAX	0.460

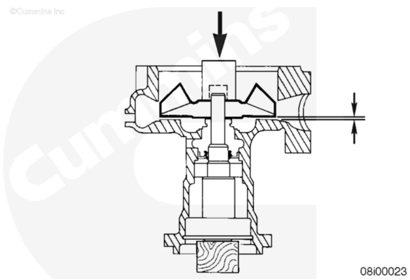
Install one drop of Loctite® 290, or equivalent, as shown.

Apply a smooth coating of Loctite® 609, or equivalent, to the I.D. of the impeller.



Install the impeller with a mandrel and arbor press.

Press the impeller onto the shaft to the specified clearance. Use a feeler gauge through the water outlet port to measure the clearance.



Impeller Vane to Body Clearance		
mm		in
0.83	MIN	0.033
1.01	MAX	0.040

Lubricate a new o-ring (4) with vegetable oil.

Install the o-ring onto the valve (2).

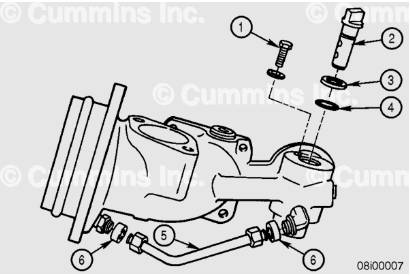
Install the valve assembly and washer (3) into the housing.

Install and tighten the capscrew (1).

Torque Value: 20 n•m [177 in-lb]

Lubricate new grommets (6) with vegetable oil.

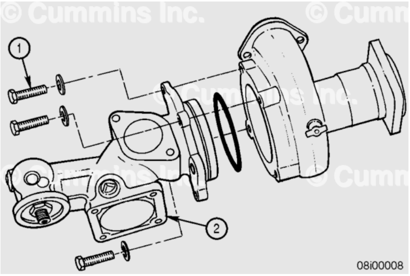
Install the grommets and transfer tube (5) onto the housing.



Lubricate the o-ring with vegetable oil.

Install the o-ring, inlet housing (2), and capscrews (1).

Torque Value: 45 n•m [33 ft-lb]

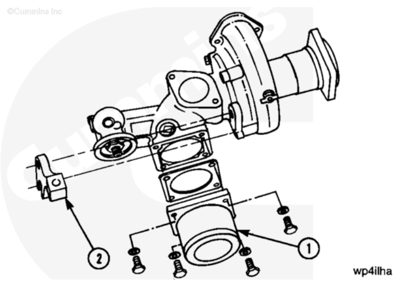


Install the gasket, inlet connection (1), support bracket (2), and capscrews.

Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]

Do **not** tighten the support bracket until the water pump is assembled to the engine. Use a heavy, plain washer on the capscrew that attaches through the slotted hole.



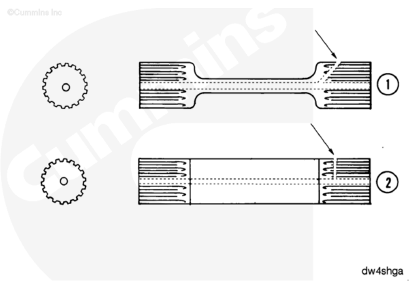
Install

Failure of the water pump drive shaft will result if the wrong shaft is used.

The phenolic (plastic) impeller has been discontinued. It is no longer available for production or service. The shaft (2) is also no longer available for production or service.

Use shaft (1) with cast iron impeller pumps. Use shaft (2) with phenolic impeller pumps.

Install the shaft with the oil hole at the side of the shaft toward the water pump.



Lubricate the shaft with clean engine oil.

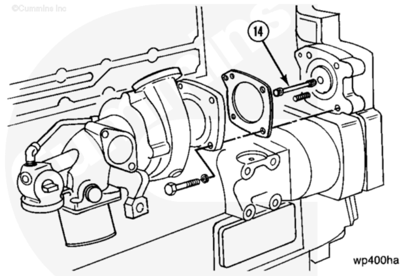
Install the shaft in the splined hole in the water pump drive.

Do **not** tighten the capscrews and nut until the support bracket is aligned with the cylinder block.

Install the gasket and water pump.

Install the three capscrews and nuts.

Tighten the capscrews and nuts.



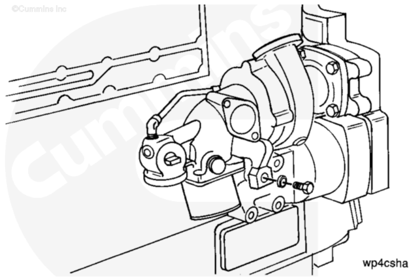
Torque Value: 45 n•m [33 ft-lb]

The support bracket **must** be flat against the water pump and the cylinder block.

Align the bracket before tightening the capscrews.

Install the capscrew through the bracket.

Tighten the capscrew.



Torque Value: 206 n•m [150 ft-lb]

Check the alignment between the support bracket and the water pump.

Loosen the capscrew and adjust, if necessary.

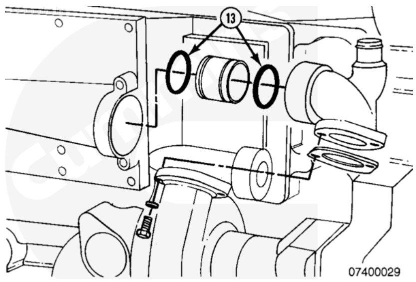
Tighten the capscrews holding the support bracket to the water pump.

Torque Value: 45 n•m [33 ft-lb]

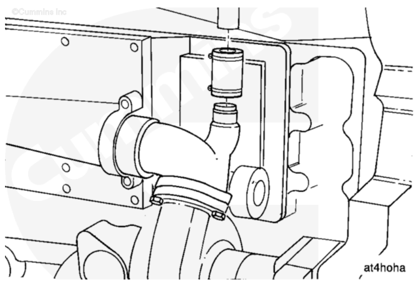
Lubricate the o-rings (13) with vegetable oil and install them onto the transfer tube.

Install the transfer tube into the water outlet connection.

Install the water outlet connection assembly.

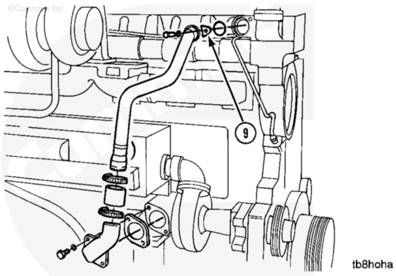


Connect the water pump outlet connection to the aftercooler supply hose.



Install the water bypass tube.

Install the water bypass tube clamp and capscrew.



Finishing Steps

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the coolant filter. Refer to Procedure 008-006 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-006-tr.html).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html).
- Connect the batteries or air starter supply line. See equipment manufacturer service information.
- Operate the engine and check for leaks.

Last Modified: 09-Jul-2015